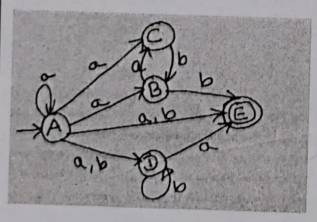
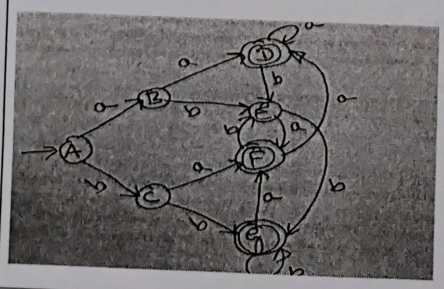
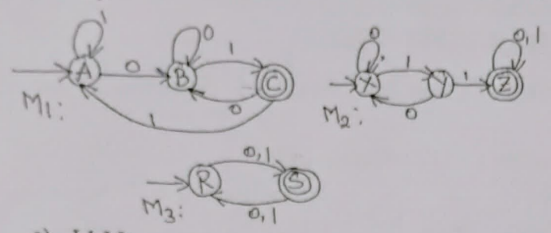
	R V College of Engineering **Department of Computer Science and Engineering** **TEST-I: Question Paper**		
Course : (Code)	**THEORY OF COMPUTATION (21CS54)**	**Semester : 5th**	
Date: JAN-2024	**Duration : 110 minutes**	**Staff: SMS/HKK/ASP**	
Name :	**USN :**	**Section :**	**A, B, C**

1.a	Define regular expression. Give regular expressions which generates the following languages over the alphabet $\sum=\{0, 1\}$. i. Set of all strings that do not end with 00. ii. Set of all strings that do not contain the substring 01.	6	L3	C03
1.b	Answer the following questions a) Let M1 and M2 are the two NFA- ϵ. In the construction of M_u (Union of M1 and M2) consider this alternative construction: Instead of a new state q_u and ϵ-transition from it to q1 and q2, make q1 the initial state of the new NFA- ϵ, and create a ϵ-transition from it to q2. Either prove that this works in general, or give an example in which it fails. b) In the construction of M_c (Concatenation), consider the simplified case in which M1 has only one accepting. Suppose that we eliminate the ϵ-transition from the accepting state of M1 to q2, and merge these two states into one. Either show that this would always works or give an example in which it fails.	4	L3	CO4
2.a	Prove that for every NFA there exists a DFA accepting the same language. For the NFA shown below, using the subset construction draw the equivalent DFA. 	5	L4	CO4
2.b	For the DFA shown below, use the minimization algorithm to find a minimum state DFA recognizing the same language. 	5	L4	CO3

3.a	Let M1, M2 and M3 are the DFA's pictured below recognizing languages L1, L2 and L3 respectively.  a) L1-L2 b) L1∩L3 c) L2 U L3	6	L4	CO3
3.b	State and prove pumping lemma for regular languages. Use pumping lemma to show that the following language is not regular. L={x0x | x ϵ{0,1}* }.	4	L2	CO2
4.a	Define CFG. Find CFGs to generate each of the following languages. i. L={$a^i b^j c^k$ | j=i+k} ii. L={ $a^i b^j c^k$ | j=i or j=k}	6	L3	CO2
4.b	Describe the language generated by each of the following grammars. i. The grammar with the productions set {S→aA|bC|b, A→aS|bB, B→aC|bA|a, C→aB|bS} ii. The grammar with the productions set {S→bS|aA| ϵ, A→aA|bB|b, B→bS}	4	L2	CO1
5.a	Let G be the CFG with productions set as {S→S+S|S-S|S*S|S/S|(S)|a}. Answer the following w.r.t a+(a*a)/a-a. i. Give 2 left most derivations. ii. Draw the derivation tree corresponds to each of the derivations in (i). iii. How many distinct leftmost derivations are there?	5	L3	CO2
5.b	Define GNF grammar. Given below a CFG G, find a CFG G' in GNF generating L(G) – { ϵ }. S→AaA|CA|BaB, A→aaBa|CDA|aa|DC, B→bB|bAB|bb|aS, C→Ca|bC|D, D→bD|ε.	5	L4	CO3

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	R V College of Engineering **Department of Computer Science and Engineering** **TEST-II: Question Paper**		
Course : (Code)	**THEORY OF COMPUTATION (21CS54)**	**Semester : 5th**	
Date:FEB-2024	**Duration : 110 minutes**	**Staff:SMS/HKK/ASP**	
Name:	**USN:**	**Section:**	**A, B, C**

1.a	Give the left linear grammar to generate the language L={w | w ϵ {a,b}* and w not contain the substring ab.}. Give the derivation for the string bbaaa.	4	L4	CO3
1.b	Define PDA and the language accepted by PDA. Construct PDA for L={w| w ϵ {a,b}* and w is palindrome}. Show by IDs the string abaaba is accepted by the PDA constructed.	6	L3	CO3
2.a	Define DPDA. Give DPDA to accept $L=\{a^n b^{2n} \mid n>0\}$. Illustrate the operation of the constructed DPDA on the input string aaabbbbbb.	4	L3	CO4
2.b	What are steps to be followed while finding an equivalent PDA by empty stack for the given CFG. Find the equivalent PDA to the CFG with productions S→aSb | bSa | abS | baS | Sab | Sba | ϵ. Show that the string abaabb is generated by the given CFG and it is accepted by its equivalent PDA.	6	L4	CO4
3.a	Show that every regular language is a deterministic context free language.	2	L1	CO1
3.b	Let L be L(M1) for some PDA with empty stack $M1=(Q,\Sigma,\Gamma,\delta,q_0,Z_0,\phi)$, prove that there exists an final state PDA M2 such that L=L(M2).	2	L2	CO1
3.c	Let $L1=\{a^i b^j c^k \mid i= j\}$ and $L2=\{a^i b^j c^k \mid j= k\}$. Show that L1 and L2 are context free but L3=L1∩ L2 is not context free.	6	L2	CO2
4.	Define Linear Bounded Automata, Language acceptance by Linear Bounded Automata. Design Linear Bounded Automata to accept $L=\{ ww^R \mid w \in \{a, b\}^* \}$. Using Instantaneous Descriptions show that the string *babbab* is accepted by the Linear Bounded Automata.	10	L4	CO4
5.a	Design Turing Machine to perform the string concatenation operation. f(x,y)=xy, where x, y ε {a, b, c}*. Use Instantaneous Descriptions to show operation on x=*abc and y=bca.*	6	L3	CO4
5.b	Define the Unrestricted Grammar. Consider the unrestricted grammar with the given productions S-->TDE, T-->ABCT |∈, AB-->BA, BA-->AB, CA-->AC, CB-->BC, CD-->DC, CE-->Ea, BD-->Db, A-->a, D-->∈, E-->∈. Describe the language generated by this grammar.	4	L4	CO3

	R V College of Engineering **Department of Computer Science and Engineering** **TEST-III: Question Paper**
Course: (Code)	**THEORY OF COMPUTATION (21CS54)** — **Semester: 5th**
Date: MAR-2024	**Duration: 110 minutes** — **Staff: SMS/HKK/ASP**
Name:	**USN:** — **Section:** **A, B, C**

1.	Define Linear Bounded Automata, Language acceptance by Linear Bounded Automata. Design Linear Bounded Automata to accept $L=\{w \mid w \in \{a, b, c\}^* \text{ and } N_a(W)=N_b(W)=N_c(W)\}$. Using Instantaneous Descriptions show that the string *baccba* is accepted by the constructed Linear Bounded Automata.	10	L4	CO4
2.a	Define Context Sensitive Grammar (CSG), Language generated by CSG. Give CSG to generate the language $L=\{a^n b^n a^n \mid n\geq 0\}$. Show the derivation for the string aaabbbaaa.	6	L3	CO3
2.b	Define Post Correspondence Problem (PCP). Solve the instance of the below PCP.	4	L2	CO2

	List X	List Y
i	Xi	Yi
1	1	10
2	0	10
3	010	01
4	11	1

3.a	Define Unrestricted Grammar (URG), Language generated by URG. Give URG to generate the language $L=\{a^n b^n c^n d^n \mid n \geq 0\}$. Show the derivation for the string aaabbbcccddd.	6	L3	CO3
3.b	Explain Chomsky hierarchy in detail.	4	L1	CO1
4.	Define Turing Machine (TM), Language of TM. Design the TM to accept the language $L=\{ww \mid w \in \{a, b\}^*\}$. Trace the operation of the constructed TM on the string baabaa.	10	L4	CO4
5.a	Design Turing Machine to perform the string reverse operation over $\Sigma=\{a, b, c\}$. Use Instantaneous Descriptions to show operation on *w=abac,*	6	L4	CO4
5.b	Let L_1 and L_2 are recursively enumerable languages. Show that $L_1 \cup L_2$ and $L_1 \cap L_2$ are also recursively enumerable languages.	4	L3	CO3